

QCE Chemistry Units 3&4

Test 3: Oxidation and reduction

- Redox reactions
- Electrochemical cells

SECTION 1 – MULTIPLE CHOICE QUESTIONS

QUESTION 1 B

B is correct. A redox reaction involves a change in oxidation state. In this option, all the substances maintain the same oxidation state throughout the reaction.

A is incorrect. Mg is oxidised (oxidation state changes from 0 to +2) and the hydrogen ions (H^+) are reduced (+1 to 0).

C is incorrect. C is oxidised (0 to +4) and O_2 is reduced (0 to -2).

D is incorrect. Fe is oxidised (0 to +3) and O_2 is reduced (0 to -2).

QUESTION 2 B

B is correct. In this compound, the iron ion has a charge of 3+ and, therefore, has an oxidation state of +3.

A is incorrect. Al is an element, so it has an oxidation state of 0.

C is incorrect. In this compound, the sodium ion has a charge of 1+ and an oxidation state of +1.

D is incorrect. In this compound, the iron ion has a charge of 2+ and an oxidation state of +2.

QUESTION 3 C

C is correct and **D** is incorrect. Oxidation involves the loss of electrons. Magnesium has two electrons in its outer shell and will, therefore, form a 2+ ion.

A and **B** are incorrect. These half-equations show a gain of electrons.

QUESTION 4 A

A is correct and **B** is incorrect. Oxidation involves the loss of electrons and always occurs at the anode. Therefore, electrons move from the anode to the cathode and cannot move from the cathode to the anode.

C and **D** are incorrect. Ions, not electrons, move through the salt bridge.

QUESTION 5 C

C is correct and **B** is incorrect. Reduction involves the gain of electrons and oxidation involves the loss of electrons.

A and **D** are incorrect. Both an increase in the oxidation state and a loss of hydrogen indicate oxidation, not reduction.

QUESTION 6 B

Atoms in group 7 (halogens) are more likely to gain electrons than lose them, so they will have a negative oxidation state. Halogens only require one extra electron to complete their valence shell. Therefore, they are likely to have an oxidation state of -1 .

QUESTION 7 B

B is correct. The reducing agent is the substance that undergoes oxidation. In the equation, the oxidation state of sulfur (S) changes from $+4$ to $+6$, indicating that oxidation has occurred. SO_3^{2-} is, therefore, the reducing agent.

A is incorrect. MnO_4^- is reduced and is, therefore, the oxidising agent.

C and **D** are incorrect. These options are the products of the reaction and, therefore, do not cause oxidation or reduction to occur.

QUESTION 8 C

C is correct. Hydrogen only has an oxidation state of -1 when it is bonded to a metal to form a metal hydride.

A, **B** and **D** are incorrect. These substances have a hydrogen atom with an oxidation state of $+1$.

QUESTION 9 B

In hydrogen peroxide (H_2O_2), hydrogen has an oxidation state of $+1$, meaning that the oxidation state of oxygen must be -1 so that the molecule is 0 overall.

QUESTION 10 C

In the reaction, the oxidation state of zinc (Zn) changes from 0 to $+2$, indicating oxidation. The oxidation state of hydrogen (H) changes from $+1$ to 0, indicating reduction.

SECTION 2**QUESTION 11 (1 mark)**

When oxidation occurs, the electrons that are lost by the substance being oxidised must be accepted by other atoms or ions via reduction.

[1 mark]

1 mark for stating that the electrons that are lost must be accepted by other atoms or ions.

QUESTION 12 (3 marks)

For example, any three of:

Electrolytic cell	Galvanic cell
involves a non-spontaneous redox reaction	involves a spontaneous redox reaction
has a positively charged anode and a negatively charged cathode	has a negatively charged anode and a positively charged cathode
can occur in the same beaker	requires the half-cells to be separated
requires an external power source	does not require an external power source
converts electrical energy to chemical energy	converts chemical energy to electrical energy
does not require a salt bridge	requires a salt bridge

[3 marks]

1 mark for each difference identified.

QUESTION 13 (1 mark)

$$\text{Cl} = -1$$

$$\text{Co} + 4 \times \text{Cl} = -2$$

$$\text{Co} + (-4) = -2$$

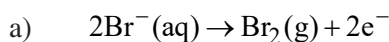
$$\text{Co} = -2 + 4$$

$$= +2$$

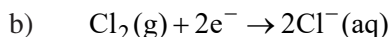
[1 mark]

1 mark for stating that the oxidation state of cobalt is +2.

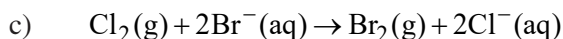
Note: Calculations are not required to obtain full marks.

QUESTION 14 (5 marks)

[1 mark]

*1 mark for writing the balanced oxidation half-equation.**Note: States are not required to obtain full marks.*

[1 mark]

*1 mark for writing the balanced reduction half-equation.**Note: States are not required to obtain full marks.*

[2 marks]

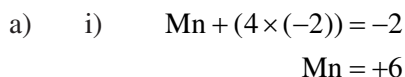
*1 mark for combining the half-equations.**1 mark for removing the electrons from the equation.**Note: States of matter are not required to obtain full marks.*

[1 mark]

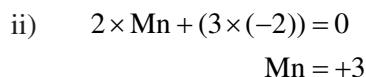
*1 mark for identifying the oxidising agent in the reaction.***QUESTION 15 (2 marks)**

metal C < metal A < metal B

[2 marks]

*1 mark for listing 1–2 metals in the correct positions.**2 marks for listing all three metals in the correct positions.***QUESTION 16 (4 marks)**

[1 mark]

1 mark for determining the oxidation number of MnO_4^{2-} .

[1 mark]

1 mark for determining the oxidation number of Mn_2O_3 .As MnO_4^{2-} has a higher oxidation number than Mn_2O_3 , it is more likely to gain electrons and be reduced, which means it is a stronger oxidising agent.

[2 marks]

*1 mark for determining that MnO_4^{2-} would be the stronger oxidising agent.**1 mark for justifying that the MnO_4^{2-} is the stronger oxidising agent due to its higher oxidation number.**Note: Consequential on answers to Questions 16ai) and aii).*

QUESTION 17 (6 marks)

- a)
- $N_2 = 0$
- and N in
- $NO_2 = +4$

The increase in the oxidation state indicates that N_2 is undergoing oxidation.

- $O_2 = 0$
- and O in
- $NO_2 = -2$

The decrease in the oxidation state indicates that O_2 is undergoing reduction.

[4 marks]

1 mark for determining the change in the oxidation state of N_2 .

1 mark for determining that N_2 is undergoing oxidation based on the increase in the oxidation state.

1 mark for determining the change in the oxidation state of O_2 .

1 mark for determining that O_2 is undergoing reduction based on the decrease in the oxidation state.

- b)
- N_2
- is the reducing agent and
- O_2
- is the oxidising agent.

[2 marks]

1 mark for identifying that N_2 is the reducing agent.

1 mark for identifying that O_2 is the oxidising agent.

QUESTION 18 (6 marks)

half-equation 1: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

Multiplying half-equation 1 by 2 gives $2MnO_4^- + 16H^+ + 10e^- \rightarrow 2Mn^{2+} + 8H_2O$.

half-equation 2: $2Cl^- \rightarrow Cl_2 + 2e^-$

Multiplying half-equation 2 by 5 gives $10Cl^- \rightarrow 5Cl_2 + 10e^-$.

$2MnO_4^- + 16H^+ + 10e^- + 10Cl^- \rightarrow 2Mn^{2+} + 8H_2O + 5Cl_2 + 10e^-$

$2MnO_4^- + 16H^+ + 10Cl^- \rightarrow 2Mn^{2+} + 8H_2O + 5Cl_2$

[6 marks]

1 mark for determining half-equation 1 to show the reduction of MnO_4^- .

1 mark for balancing half-equation 1 by multiplying by 2.

1 mark for determining half-equation 2 to show the oxidation of Cl^- .

1 mark for balancing half-equation 2 by multiplying by 5.

1 mark for combining the half-equations.

1 mark for removing the electrons from the equation.

QUESTION 19 (2 marks)

The copper container was not oxidised because copper is not oxidised in the presence of zinc ions.

Therefore, no change was observed.

The zinc container degraded because zinc is oxidised in the presence of copper ions. Therefore, the container begins to break down, causing the copper chloride solution to spill out of the container.

[2 marks]

1 mark for explaining why the copper container was unchanged.

1 mark for explaining why the zinc container degraded.